

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



443/1

Agriculture

Paper 1

Time: 2 Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO STUDENTS

- (a) Write your name and index number in the spaces provided at the top of this page.
- (b) Sign and write the date of the examination in the spaces provided.
- (c) This paper consists of three sections: **A**, **B** and **C**.
- (d) Answer all questions in sections **A** and **B**.
- (e) Answer any two questions in section **C**.
- (f) All answers should be written in the spaces provided.

FOR EXAMINER'S USE ONLY

SECTION	QUESTIONS	MAXIMUM SCORE	CANDIDATES SCORE
A	1 – 19	30	
B	20– 23	20	
C		20	
		20	
	TOTAL SCORE	90	

SECTION A- (30 MARKS)

Answer all questions in the spaces provided

- 1) State the **three** forms in which horticulture is practiced in Kenya (1½ marks)

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- 2) Identify **two** ways in which agriculture support agro based industries (1 mark)

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- 3) List down **three** biotic factors that influence agriculture negatively (1½marks)

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- 4) Name any **three** crops that can be propagated by use of suckers (1½marks)

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- 5) Outline **four** roles of trees in soil conservation (2 marks)

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- 6) Identify **two** examples of intensive farming systems practiced in crop production (1 mark)

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7) Outline **four** ways in which living organisms contribute to soil formation (2 marks)

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8) Give **four** reasons that may necessitate sub soiling to be carried out on a piece of land (2 mks)

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9) Name any **four** physical methods of water treatment (2 marks)

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10) Give **three** reasons why green manure is not commonly used (1½marks)

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11) Name **two** types of labour records that can be kept by a large scale plantation farmer.(1 mark)

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12) Outline **four** areas that should be avoided when collecting a soil sample for testing (2 marks)

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13) Differentiate between a seed bed and a seedling bed (1 mark)

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14) Name **three** field practices that are used to support weak stemmed crops (1½ marks)

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15) Give **four** reasons that would make a farmer choose a jembe during land preparation over a disc plough (2 marks)

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16) Give a reason why use of organic manure is discouraged in carrot production (1 mark)

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17) Outline **four** advantages of zero grazing as a system of grazing (2 marks)

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18) Identify **four** sources of credit facilities where one can get capital to carry out crop production (2 marks)

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19) Name any **three** fertilizer elements (1½ marks)
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SECTION B (20 MARKS)

Answer all questions in this section.

20) The following are photographs of weeds obtained from the farm



i) Identify the weeds (2 marks)
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Price (ksh)	Quantity demanded (kgs)
10	250
9	350
8	550
7	600
6	700
5	1000
4	1500

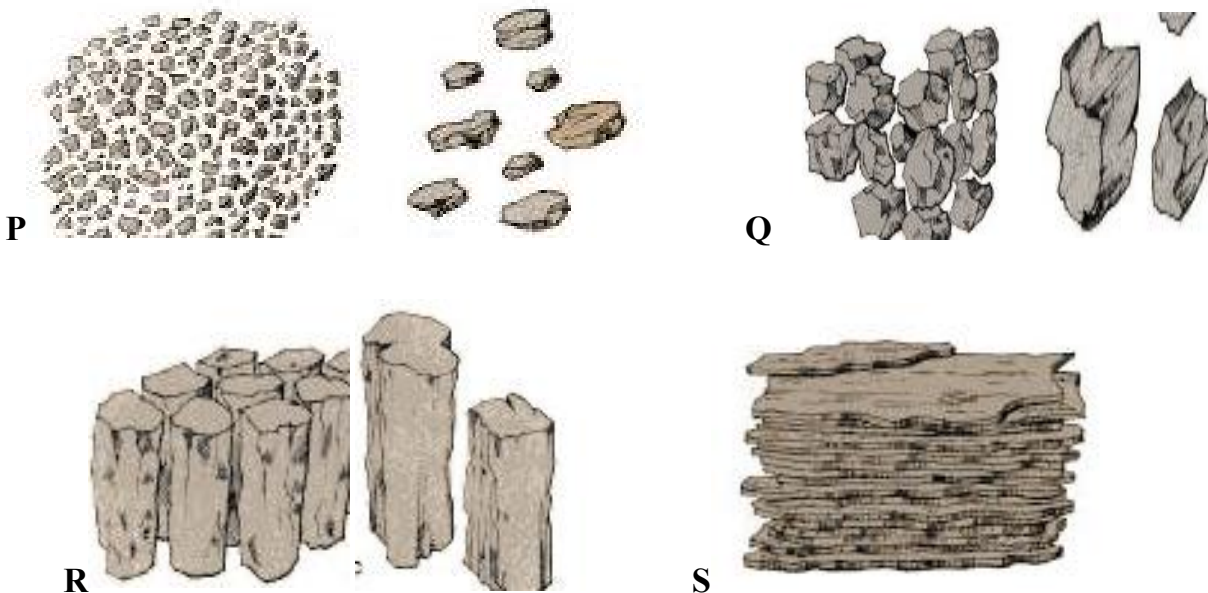
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b) Other than price state any other factor that may factor that may increase the demand for rice in the market (2 marks)

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22)The following diagrams shows various samples of soil aggregates



a) Identify the type of soils structures (2marks)

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b) Briefly explain how soil structures P can be improved (1mark)

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c) Name two places where soil structure labeled S can be found (2 marks)

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23) Study the following photographs and use them to answer the question that follows



a) Identify the type of irrigation shown in the photograph (1 mark)

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b) State any **two** advantages of the above method of irrigation (2 marks)

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c) Other than the one shown above identify any other **two** methods that a farmer can use to irrigate his crops (2 marks)

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SECTION C – 40 MARKS)

Answer any two questions and answer them in the spaces provided after question 26

- 24)** a) A farmer has 30 hectares of arable land, 20 hectares of which is planted with maize and 10 hectares with grass hay. He wishes to replace 10 hectares of maize with Irish potatoes next year. The fertilizer rate will have to be increased from 2 bags per hectare for maize to 4 bags per hectare for potatoes. As a result of the change extra 40 men days of casual labour per hectare will be necessary at the rate of Kshs.100 per man a day. The average yield per hectare of maize is 35bags and for potatoes 120bags. The price paid is Kshs.1, 000 per bag of maize and Kshs.600 per bag of potatoes. Maize seeds cost Kshs.750 per hectare and potatoes cost Kshs.800 per hectare. Fertilizer costs are Kshs. 650 per hectare.
- i) Draw up a partial budget and advice the farmer accordingly. **(9 marks)**
- ii) Name the other type of budget that can be prepared by a farm **manager** **(1 mark)**
- b) Discuss the various problems associated with land fragmentation **(6 marks)**
- c) Explain **four** environmental factors that affect the effectiveness of herbicides **(4 marks)**
- 25. a)** Discuss the various cultural practices carried by farmers to control pests **(10marks)**
- b) Discuss the production of dry beans under the following headings
- (i) Varieties **(2 marks)**
- (ii) Selection of planting materials **(3 marks)**
- (iii) Postharvest practices of dry beans **(5 marks)**
- 26 a)** Describe harvesting of pyrethrum under the following subheading
- i) Method and procedure **(3 marks)**
- ii) Precautions **(3 marks)**
- b) State **five** characteristics of nitrogenous fertilizers **(4marks)**
- c) Explain **five** factors to consider when designing a crop rotation programme **(10 marks)**